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1 Music Player SSI

The music player has two buttons marked with the numbers 1 and 0.

When 1 is pressed, the player turns on and music starts playing. If the player is turned on and 1 is pressed again, the music continues to play without interruption. The player does not turn on again.

When 0 is pressed, the player turns off and the music stops playing.

At the beginning of the day, the player was turned off. At the end of the day, this sequence was in its memory:

101100101110010

How many times has the player been turned on during the day?

- A) 8
- B) 5
- C) 15
- D) 7



Solution

The correct answer is (B).

This is Computer Science!

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2 Ambika's Message SSI

Ambika sends messages to her friends using a variety of stickers. There are two types of stickers: Animal Stickers and Special Stickers.

Sticker Types:

- Animal Stickers: Dog, Cat, and Rabbit.
- Special Stickers: Star, Heart, and Sun.

Ambika follows specific rules for adding special stickers to the end of her messages based on the animal stickers she uses:

- If a message contains exactly two animal stickers, Ambika adds a Star sticker at the end.
- If a message contains exactly three animal stickers, Ambika adds a Heart sticker at the end.
- If all the animal stickers in a message are different from each other, Ambika adds a Sun sticker at the end.

Multiple stickers can be applied if multiple rules apply to a message.

There are four different messages. Based on Ambika's rules, select the message that was correctly sent by Ambika.

- A) Message 1: Dog, Cat
- B) Message 2: Dog, Dog, Cat, Heart
- C) Message 3: Rabbit, Dog, Sun, Star
- D) Message 4: Dog, Cat, Rabbit, Heart



Solution

The correct answer is Message 2.

Message 2 is correct because it contains exactly three animal stickers (Dog, Dog, Cat), so the Heart Rule applies. However, because two of the animals are the same (two Dogs), they are not all different, so the Sun Rule does not apply. The message correctly ends with only a Heart.

Why the other messages are incorrect:

- Message 1 has two different animals, so it is missing both a Star and a Sun sticker.
- Message 3 has two different animals, so it should have both a Star and a Sun sticker at the end. (The source explanation notes that if Ambika sent this, it would require both stickers).
- Message 4 has three different animals, so it is missing a Sun sticker at the end.

Going over all the rules, it follows that only the second message could have been sent by Ambika but the first, third, and fourth messages could not.

This is Computer Science!

This task's solution follows a simple algorithm to determine which messages were sent by Ambika, which consisted of successively testing each of Amelia's rules against the set of messages and collecting the set of messages that failed any of the rules. This implementation is an example of the use of if-then conditional statements in programming.

More generally, this task is an example of a data validation process, in which some inputs are analyzed to ensure that they conform to some pre-existing rules. In particular, the "stickers" attached to each input resemble the concept of a checksum used to verify the message's integrity.

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3 Book Recommendation SSI

An app uses a rule to suggest new books to read:

If another person has read at least 2 of the same books as you, the app suggests their other books to you (if you have not read them).

You have recently read these three books:

Malgudi Days, Sherlock Holmes, Birbal

Here is what three other people have read

Anu: Sherlock Holmes, Birbal, Kalam Bela: Malgudi Days, Panchatantra, Birbal Charu: Sherlock Holmes, Panchatantra, Kabuliwala

Based on the rule, which books will the app suggest to you to read?

- A) Malgudi Days and Panchatantra
- B) Panchatantra and Kalam
- C) Sherlock Holmes and Kalam
- D) Kalam and Kabuliwala



Solution

Correct answer is B.

The suggested books to read next would be Panchatantra and Kalam

Anu read two books that you also read (Sherlock Holmes and Birbal). She meets the rule! So, the app recommends her other book: Kalam.

Bela also read two books you read (Malgudi Days, Birbal). She meets the rule too! The app recommends: Panchatantra.

Charu only read one read you have read (Sherlock Holmes). She doesn't meet the "at least 2" rule, so we don't get her books

This is Computer Science!

This task shows how a device such as a Kindle uses either Amazon recommendation engine or the Goodreads app to suggest the next book for you to read. This is called collaborative filtering.

The computer looks for "similar persons" who like the same things as you. If they have read something you haven't read yet, the app suggests it!

While this is great to find things you like, it can also trap you in a filter bubble. This means the app only shows you things you already know, and you might miss out on completely new and different ideas.

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4 Beavers Cafeteria SSI

The children came to the dining room always placed their name badge at the entrance. When leaving, they would collect their name badge from the entrance.

Every 10 minutes a child came in and every 10 minutes, a child left the dining room.

Children came to the entrance in the following order:

Adan, Bala, Charu, Dev.

Children had left in the following order:

Dev, Bala, Charu, Adan.

Which child spent the most time in the dining room?

- A) Adan
- B) Bala
- C) Charu
- D) Dev



Solution

A is the correct option.

This is Computer Science!

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5 Laptops SSI

Four students use laptops in their class. At the end of class, they always store their laptops in two boxes: Box 1 and Box 2. Each box can store two laptops each. They place the laptops carefully so that the next day, each student can take their own laptop directly without having to move or rearrange any of the others.

Today, they left the room in this order:

Ali, Beena, Chiru, Dev

Based on their timetables, they will arrive tomorrow in this order:

Beena, Ali, Dev, Chiru.

How did they arrange their laptops when they left the room?

- A) Box1 - Top - Chiru, Box1 - Bottom - Ali, Box2 - Top - Dev, Box2 - Bottom - Beena
- B) Box1 - Top - Beena, Box1 - Bottom - Ali, Box2 - Top - Dev, Box2 - Bottom - Chiru
- C) Box1 - Top - Beena, Box1 - Bottom - Chiru, Box2 - Top - Ali, Box2 - Bottom - Dev
- D) Box1 - Top - Dev, Box1 - Bottom - Beena, Box2 - Top - Chiru, Box2 - Bottom - Ali



Solution

B) is correct.

Beena is the first to enter the room. Since her laptop must be on the top of one of the two boxes, we can immediately discard (A) and (D). The next pupil to enter the room is Ali. He can pick his laptop from the bottom of the green box (C) or from the top of the blue box (B). In this step, we cannot rule out either of the two remaining options. Then, it is the turn of Dev. In either case, he can grab his laptop from the blue box: In (B), the laptop is the one on the top, while in (C) his laptop is the one on the bottom. Finally, Chiru can pick her laptop from the bottom of the green box (C) or from the bottom of the blue one (B). In these two last steps, we were not able to discard any of the two candidate solutions (B) and (C).

To resolve the issue, we need a tiebreaker between (B) and (C). To this end, we need to consider the order in which the pupils left the room. Ali was the first to leave. Therefore, his laptop must be on the bottom of either the green or of the blue box. This requirement is satisfied in option (B) but not in option (C). The correct answer is (B).

This is Computer Science!

In this Bebras task pupils put their laptops into two boxes. Laptops that were placed on top of a box last, can be grabbed first. In Computer Science, we call this kind of box a stack. Stacks are one of the most useful data structures when it comes to temporarily store data during a computation. Their underlying principle is called LIFO (last in, first out).

When using a stack, a program can perform two actions: it can push an element at the top of the stack, or it can pop the element that currently is on the top of the stack.

- Dev is the last to put his laptop in the box.
- Dev is the first who has to grab his laptop from the top of the stack. Beena has to wait until Chiru also grabbed her laptop before she can access hers.

This means that only the element at the top of the stack can be directly accessed at any given time. In order to access an element other than the one on the top, all the elements between it and the top of the stack must first be removed.

When working with exactly one stack, we can only pop elements in the reverse order of the push actions. With two stacks with two places each, four elements can be distributed among the stacks in six different ways (see picture at the end of answer explanation section).

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6 Badri's Walk SSI

Badri took a walk through the park and visited five locations: bridge, cave, forest, statute and waterfall.

Each location has an ID made from the two digits 0 and 1 as indicated in the following table.

Location ID Codes

Location	ID
bridge	000
cave	001
forest	0100
statue	1001
waterfall	011000

As he walked, Badri recorded the IDs below, from left to right, in the order he passed the location is:

00101001001011000000

Badri did not pass any location more than once.

In which order did Badri visit the five locations?

- a) bridge, cave, forest, statute, waterfall
- b) cave, forest, statute, waterfall, bridge
- c) forest, statute, waterfall, bridge, cave
- d) statute, waterfall, bridge, cave, forest
- e) waterfall, bridge, cave, forest, statute



Solution

Correct Answer: B

cave, forest, statute, waterfall, bridge

Based on what Badri recorded:

00101001001011000000

The route that he passed would be:

Each location is identified by an ID with different symbols. Let us check Badri's record of his walk.

00101001001011000000

The first three symbols **001** only match the ID of the cave. None of the other IDs start with this sequence. Therefore, the cave must be the first site Badri passed by.

00101001001011000000

The second symbol in the record starts with 0. Only IDs: **000**, **0100** and **011000** start with 0 and only **0100** matches the sequence in Badri's record. So the forest came next on Badri's walk.

001**0100**1001011000000

The third site on Badri's walk is the statute **1001** as it is the only ID that starts with a 1

0010100**1001**011000000

The fourth location is the waterfall **011000** because it is the only one that starts with 01

0010100100**011000**000

The final location is the bridge **000**. This shares the same start as the previous stop (**010**). Since we already have the fountain, only the trees are left with this start.

00101001001011000**000**

The walk finally ended at the bridge **000**, as there are only three characters left and they fit the bridge's ID.

Therefore, Badri visits the five locations in the following order: cave, forest, statute, waterfall, and bridge.

None of the other suggested sequences are correct as none of them have the cave to be the first location to be visited by Badri.



This is Computer Science!

You have just decoded a sequence of code words from a continuous code string without separators between the code words. The decoding was possible mainly because the code (the set of site IDs) in this Bebras task is prefix-free: No codeword is a prefix of any other (meaning that no codeword starts with any other codeword).

Prefix-free codes play a crucial role in informatics, mainly for data compression. Their defining property guarantees unique decodability: When a stream of code symbols – in computing, code symbols are bits very often – is read sequentially from left to right, the boundaries between codewords are unambiguous. At every position in the stream, it is clear where one code word ends and the next begins, without the need for separators or lookahead.

A well-known example for generating prefix-free codes is Huffman coding, developed by David A. Huffman in 1952. This widely used compression method assigns shorter codewords to more frequent symbols and longer codewords to less frequent ones. Huffman codes are specifically constructed to be prefix-free, ensuring that compressed data can be decoded reliably and efficiently.

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7 Homework SSI

Sara does her homework following these rules:

- she does one assignment at a time;
- at each step, she chooses the unfinished assignment that takes the shortest time;
- after completing an assignment, she takes a break for 5 minutes.

The table below shows all the assignments she has to complete and the time needed to complete each assignment

Assignment table

Assignment	Time needed
Maths	30 min
English	20 min
Science	45 min
Music	25 min
Computing	60 min

50 minutes after starting the first assignment, Sara added a 40-minute dog walk to the list of assignments.

In which order will Sara complete all of her assignments?

- A) Maths, English, Science, Music, Computing, walking the dog
- B) English, Music, Maths, walking the dog, Science, Computing
- C) Computing, Science, walking the dog, Maths, Music, English
- D) English, Music, walking the dog, Maths, Science, Computing



Solution

The correct answer is D)

The correct answer is English, Music, Math, walking the dog, Science, Computing.

Sara started with the assignment which requires the shortest amount of time to complete it, which is English. She took 20 minutes to finish it.

Then she did the second shortest assignment, that is music, which took her 25 minutes.

For her third assignment, Sara answered the maths assignment, which took her 30 minutes.

While doing maths, she decided she had to go for a 40-minute walk with her dog. However, she first finished her maths assignment.

Since the time for walking the dog is less than the time required for the remaining assignments (Science and Computing), Sara took her dog out for a walk as the fourth assignment.

Next Sara answered her science assignment, which took her 45 minutes.

Finally Sara answered her computing assignment, which took her 60 minutes.

The other options are incorrect for the following reasons.

Maths, English, Science, Music, Computing, walking the dog is an incorrect option as the first homework assignment to be completed Maths is not the one that requires the shortest amount of time to complete it.

Computing, Science, walking the dog, Maths, Music, English is an incorrect option as the first homework assignment to be completed Maths is not the one that requires the shortest amount of time to complete it.

English, Music, walking the dog, Maths, Science, Computing is an incorrect option because the order should be English, Music, Math, and then walking the dog. So incorrect order.

English, Maths, Music, walking the dog, Science, Computing is an incorrect option as the order should be English, Music, Math, and then walking the dog. So incorrect order.

This is Computer Science!

Sara decided which assignment to do first by using the shortest job first rule: she started with the assignment that would take the least time to complete among the unfinished assignments and finish it before starting another.

Shortest Job First is a scheduling rule computers use to decide which task the CPU should do next. The CPU chooses the task with the expected shortest execution time from the list of not yet completed tasks. The tasks may be added at different times, but the new task is only chosen when the current task is completed.



Between tasks, the computer needs a very short time to save the current data and prepare the data needed for the next task.

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8 Place and Take SSI

A robot is stacking boxes of fruits in the warehouse.

The actions are as follows:

Place the fruit box on top of the stack.

Take the box from the top of the stack and put it on the cart.

The robot starts a stack and perform the following actions:

1. Place the apples
2. Place the bananas
3. Place the oranges
4. Take
5. Take
6. Place the pears
7. Place the lemons
8. Place the grapes
9. Take
10. Take

Which fruit boxes are left on the stack?

- A) pears, grapes
- B) bananas, oranges
- C) apples, pears
- D) oranges, apples



Solution

C is the correct option.

This is Computer Science!

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9 Emptying the Pantry SSI

Bhanu invited his friends to the party. He wants to bake a cake.

At home he has: flour, sugar, eggs, butter, cream, fruit

Bhanu only knows how to bake four cakes. For each of them, he has a list of the ingredients.

Sponge cake: flour, sugar, eggs, butter, milk.

Swiss roll: flour, sugar, eggs, butter, cocoa.

Fruit fizz: flour, sugar, eggs, cream, fruit.

Cheesecake: flour, sugar, butter, cottage cheese, fruit.

Which cake can Bhanu bake?

- A) Sponge cake
- B) Swiss roll
- C) Fruit fizz
- D) Cheesecake



Solution

The correct answer is C.

This is Computer Science!

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10 Garden Watering SSI

A Beaver wants to water all the trees in the garden. He only moves on the pathways.

The lengths of the pathways are:

teak – deodar: 30 meters

deodar – oak: 50 meters

sal - oak: 60 meters

deodar – sal: 20 meters

teak – sal: 50 meters

In what order should Beaver water the trees so that his route was as short as possible?

- A) teak - deodar - sal - oak
- B) teak - sal - oak - deodar
- C) teak - deodar - oak - sal
- D) teak - sal - deodar - oak



Solution

Correct Answer is A)

This is Computer Science!

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11 Box Sorter SSI

On a conveyor belt, there are T-shirts lined up in a line. Each T-shirt can be represented by a word indicating its SIZE (small, medium, large) and the NUMBER drawn on it.

For example, a small t-shirt with the number 7 will be represented as 7 small.

The robot repeatedly executes the following instructions as long as there are any T-shirts on the conveyor belt:

1. Take the first T-shirt in the line.
2. If it is small, move it to container X.
3. If it is large, move it to container Y.
4. If it is medium, check the number on the T-shirt:
 - if it is even, move the T-shirt to container X.
 - if it is odd, move the T-shirt to container Y.

Initially, container Y already contains one T-shirt and there are following T-shirts on the conveyor belt:

6 small, 2 medium, 7 small, 8 large, 4 medium, 4 small, 7 small, 3 large, 1 medium

What is the final number of T-shirts in containers X and Y?

- A) X : 6, Y : 3
- B) X : 6, Y : 4
- C) X : 7, Y : 3
- D) X : 7, Y : 2



Solution

Correct answer is B.

This is Computer Science!

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